

How Easily Could This Conclusion Be Overturned? A Framework for Robustness and Fragility Analysis of Statistical Tests in Clinical Trials

Version 2.2 — August 2026. Updated for the prospective Welch fail-closed result, the `fisher_exact` calibration, and the completed `lm_ancova` negative-result program. Software: `stabilitest` R package v0.6.0 (Ally, 2026).

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Abstract

Background: The reliability of statistical conclusions in clinical trials depends not only on p-values but on the stability of those conclusions under perturbation of the data. Regulatory guidance (ICH E9(R1)) calls for sensitivity analyses, yet standardized, quantitative frameworks for conclusion robustness remain limited.

Objective: To develop, validate, and calibrate a framework for assessing the robustness of statistical test conclusions through combined jackknife, worst-case observation removal, and bootstrap resampling.

Methods: The framework comprises: (1) jackknife leave-one-out analysis identifying influential observations; (2) a *worst-case removal analysis* — greedy adversarial deletion in the spirit of the maximum influence perturbation (Broderick, Giordano & Meager, 2023) — yielding a *removal fragility index*, a greedy upper bound on the size of a minimal overturning subset; (3) bootstrap resampling estimating the *reproducibility probability* (Goodman, 1992). A composite 0–100 score combines the components with pre-specifiable weights (default 0.4/0.4/0.2; the active Fisher calibration uses jackknife-light 0/0.5/0.5). The historical Welch simulation is retained as descriptive evidence, not as confirmatory calibration. A prospective Welch study used frozen training requirements and a sealed held-out layer; independent calibrations for `fisher_exact` (validated two-band) and `lm_ancova` / `lm_ancova_v2` / `lm_ancova_v3` (named negative results) are also reported. The framework extends to linear-model (ANCOVA), GLM, and Cox terms via a common case-deletion engine; the accompanying software also supports proportion tests, Brunner–Munzel, and TOST equivalence/non-inferiority.

Results: Type I error was preserved in clean data (0.046–0.052) and inflated by contamination (up to 0.092), which the historical Welch simulation flagged descriptively. Conditional on statistical significance, chance findings under the null averaged a robustness score of 52 with a median worst-case fragility of 1–2 observations (2% of the sample), whereas true large effects ($d = 0.8$, $n = 50$) averaged 75 with median fragility of 12–14 observations. Grand-mean-ranked removal — the usual “remove the outliers” heuristic — overstated stability dramatically (median flipping set 13–31 observations vs 2–6 under adversarial removal in the same data). In the prospective `welch-2026-2` training set, however, no candidate mapping met the frozen false-reassurance and identification requirements (`no_feasible_thresholds`); held-out validation was not opened and Welch labels were suppressed. Separately, `fisher_exact` activates a two-band Fragile / Not fragile mapping at $L = 58$ under explicit jackknife-light weights (held-out $FR \leq 0.0617$, $RI \geq 0.6767$). ANCOVA categorical-band attempts found no feasible cutoff ($v1/v2$), and the later violation-detection study was not confirmed (ΔAUC 0.0053).

Conclusions: The framework provides transparent, quantitative robustness assessment suitable for regulatory submissions, operationalizing ICH E9(R1) sensitivity-analysis principles with numeric component metrics available across methods. Fisher’s exact test is the only currently active categorical mapping, within its frozen profile and weights. Welch and ANCOVA retain numeric outputs but suppress categorical labels after prospective studies did not earn feasible mappings.

Keywords robustness analysis; sensitivity analysis; fragility index; jackknife; bootstrap; maximum influence perturbation; clinical trials

1. Introduction

1.1 Background

Statistical hypothesis testing in clinical trials typically reduces evidence to a binary decision at $p < \alpha$. This framework conveys nothing about the *stability* of the decision: a p-value of 0.048 obtained because of a single extreme responder and a p-value of 0.048 that survives any plausible perturbation of the data are treated identically. The ICH E9(R1) addendum makes robustness to assumptions and data anomalies a regulatory expectation, but provides no specific quantitative machinery for it.

Three questions, each requiring a different tool, jointly characterize the stability of a test conclusion:

1. *Which individual observations drive the result?* (influence)
2. *What is the smallest set of observations whose removal would overturn the conclusion?* (fragility)
3. *Would a replicate sample likely reach the same conclusion?* (reproducibility)

1.2 Relation to existing approaches

The fragility index of Walsh et al. (2014) counts outcome-event flips in 2×2 tables; it does not apply to continuous endpoints and has known limitations (Potter, 2020). Classical influence diagnostics (Cook, 1977; Belsley et al., 1980) quantify influence on *estimates*, not on *conclusions*. Broderick, Giordano and Meager (2023) formalized the question of the smallest data subset whose removal reverses a conclusion (the maximum influence perturbation) and showed that in prominent published analyses removing under 1% of observations can flip signs of estimated effects. Reproducibility probability was developed by Goodman (1992) and Shao & Chow (2002). The contribution of the present framework is the integration of all three perspectives into a single pre-specifiable analysis with clinically interpretable outputs.

Terminology: throughout, *removal fragility index* denotes the number of removed observations at which the conclusion first changes. It is related to but distinct from the Walsh fragility index, which flips event status without changing n .

1.3 What changed in version 2

The January 2026 version had four methodological weaknesses, documented in the accompanying review: its bootstrap “stability” metric conflated strength of evidence with robustness; its removal analysis ranked observations by distance from the grand mean, which under a genuine treatment effect preferentially removes true responders and is far from the worst case; its composite score was mis-scaled (the fragility term could not fall below 70); and its interpretation thresholds were uncalibrated. Version 2 addresses the metric and algorithmic problems, while later method-specific prospective studies determine separately whether categorical interpretation can be earned.

1.4 Current calibration policy

The runtime registry is keyed by the resolved method, endpoint, and observed conclusion (`welch_unpaired`, `paired_t`, `fisher_exact`, `lm_ancova`, and so on); it does not use the generic `two_sample` identity. Every analysis returns the numeric composite score and component metrics. Welch categorical labels are suppressed; numeric scores and component metrics remain available. Categorical labels are emitted only for a significant, applicable `fisher_exact` analysis under the explicit jackknife-light weights `fragility = 0.5`, `bootstrap = 0.5`, `jackknife = 0` (two-band Fragile / Not fragile at $L = 58$; version `fisher-2026-1`; provenance `study:binary_proportion@cc3344931614`). Default 0.4/0.4/0.2 Fisher scores remain numeric-only. Otherwise the categorical label is left blank (represented as NA in R). The public `robustness_analysis()` dispatcher and its existing test choices are unchanged. The broad-family Welch tables and 55/70 anchors are archived as historical descriptive evidence under `manuscript/calibration/published/`; they are not active calibration inputs or licensed interpretation bands. The historical `lm_ancova` calibration program (v1–v3) is retained as a named

negative-result contribution with uncalibrated registry rows; the next independent positive calibration target is `chi_square_2x2` (binary-proportion Phase 2).

2. Methods

2.1 Framework overview

Given two samples (or a model term; Section 2.6) and a test with significance level α , four analyses are performed:

Component 1 — Jackknife (leave-one-out). Each observation is deleted in turn and the test repeated (n tests), following the classical leave-one-out framework (Shao & Tu, 1995). Outputs: conclusion stability S_{jack} (% of deletions preserving the conclusion), the set of influential observations (deletion flips the conclusion or shifts p by more than $\delta = 0.05$), and the leave-one-out p -value range. Because single-deletion influence shrinks as $\sim 1/n$, high jackknife stability in large samples is *weak* evidence of robustness; the component's chief value is identifying specific observations for data-quality review.

Component 2 — Worst-case removal (primary fragility analysis). A greedy algorithm deletes, at each step, the observation whose removal moves the p -value furthest toward overturning the conclusion, up to $k_{\text{max}} = \lfloor 0.30 n \rfloor$ deletions. The *worst-case removal fragility index* k_{frag} is the step at which the conclusion first flips ($k_{\text{max}} + 1$ if it never does). The greedy set is *sufficient* to overturn the conclusion; a smaller set may exist, so k_{frag} is an upper bound on the size of the minimal overturning subset — in practice a much tighter bound than any symmetric outlier-trimming rule provides.

Component 3 — Extreme-value removal (descriptive). The $v1$ procedure — cumulative deletion of observations most distant from the (recomputed) grand mean — is retained for continuity and because the *gap* between its fragility index and the worst-case index is itself informative: it measures how much protection the “obvious outliers” narrative provides relative to an adversarial reading of the data. It does not enter the composite score.

Component 4 — Bootstrap reproducibility. B resamples (default 1000) drawn with replacement within groups (Efron & Tibshirani, 1993); S_{boot} is the proportion reaching the original conclusion. Because resampling preserves the observed effect, S_{boot} estimates the *reproducibility probability* — approximately the power at the observed effect size — and reflects strength of evidence, not resistance to contamination. A marginal p -value yields S_{boot} near 50–60% even in perfectly clean data. It is reported with that explicit framing, together with the mean, SD, and 2.5th–97.5th percentile interval of the bootstrap p -value distribution (a descriptive interval; p -values are not parameters and this is not a confidence interval).

2.2 Composite robustness score

$R = w_1 \cdot S_{\text{jack}} + w_2 \cdot F + w_3 \cdot S_{\text{boot}}$, where $F = 100 \cdot \min(k_{\text{frag}} / (k_{\text{max}} + 1), 1)$ rescales the worst-case fragility index to the full 0–100 range (in $v1$ this term was bounded below by ~ 70 , inflating all scores). Default weights $w = (0.4, 0.4, 0.2)$: influence and fragility carry equal weight; the reproducibility term is down-weighted because it largely restates the p -value. Weights are an explicit argument of the software and should be pre-specified in the SAP. The active `fisher_exact` calibration uses jackknife-light weights $(0, 0.5, 0.5)$ and does not change the default numeric output for interactive analyses.

2.3 Interpretation policy and active calibrations

Scores and component metrics are descriptive outputs for every supported method. One active categorical mapping is licensed:

1. **Fisher two-band.** Significant `fisher_exact` under explicit jackknife-light weights, a canonical two-arm profile (per-arm n in $[25, 200]$, allocation ratio in $[0.8, 1.25]$, control-arm event rate in $[0.08, 0.55]$ with ≥ 3 events and ≥ 3 non-events, $\alpha = 0.05$, $n_{\text{boot}} = 1000$), and `max_removal_pct = 0.30`: score ≤ 58 is **Fragile**, score > 58 is **Not fragile**. There is no Robust tier. Version `fisher-2026-1`; provenance `study:binary_proportion@cc3344931614`.

Welch, paired t , rank, non-canonical binary, model, Cox, and TOST methods, as well as non-significant results and unsupported weight/design configurations, retain scores and components but leave the categorical label blank (represented as NA in R).

2.3.1 Prospective Welch calibration (fail-closed)

The prospective Welch study, version `welch-2026-2`, ended at training with `no_feasible_thresholds`. Its exact calibration unit was a significant `welch_unpaired` mean-difference result at $\alpha = 0.05$, default weights (0.4, 0.4, 0.2), $n_{\text{boot}} = 1000$, and a 30% removal cap. Training comprised 4,500 completed significant analyses across null, borderline, and clear truth strata. The historical 55/70 mapping produced false reassurance of 0.458 and clear-effect identification of 0.499 in that training set, far outside the frozen requirements.

The frozen no-candidate decision has hash `9c45481b952cab7cb9b9086e37924a39d83fe0484628745dbe2e79eb33e8797d`. Because no candidate met the frozen requirements, the held-out validation data were not opened. The registry therefore marks `welch_unpaired` uncalibrated and suppresses its categorical label. This is not a failed held-out confirmation: the pre-specified training requirements stopped the study before validation access.

The historical 55/70 bands remain a retired exploratory comparator. Their descriptive separation motivated the prospective study but does not license a Robust, Moderately Robust, or Fragile verdict for a current Welch result.

2.3.2 `fisher_exact` calibration (binary-proportion Phase 1)

An independent study (`manuscript/calibration/studies/binary_proportion/`) delivered a validated two-band Fragile / Not fragile decision at cutoff $L = 58$ for eligible significant canonical two-arm Fisher exact results. Truth strata were exact-power-defined against enumerated exact Fisher power (null exact; borderline exact power 0.60 diagnostic-only; clear exact power 0.95). Training selected $L = 58$ (FR 0.0493, Wilson upper 0.0607; RI 0.7608, Wilson lower 0.74). Held-out confirmation under the conservative-of-Wilson-and-cluster-bootstrap rule yielded FR upper ≤ 0.0617 and RI lower ≥ 0.6767 . The runtime activates this decision in the runtime registry only when the analysis profile and explicit jackknife-light weights match the frozen design; default 0.4/0.4/0.2 weights never emit the band. The prospectively frozen `onc_response_trial` illustration is excluded from calibration ledgers.

2.3.3 ANCOVA calibration as a named negative result

The `lm_ancova` program is reported as a named negative-result contribution, not as a transferable Welch fallback:

- **First ANCOVA training study.** No feasible thresholds were found (`no_feasible_thresholds`; candidate hash `9ccfc2fca7c0a07c19a3a18838e9a3f2`; held-out not opened; version `lm-ancova-2026-1`).
- **Jackknife-light two-band ANCOVA study.** This second attempt likewise stopped after training after a sealed pilot that returned a false-GO on location metrics ($\Delta = 24.4$, overlap = 0.034, AUC = 0.892) while training found no feasible L (best RI 0.554 < 0.70; candidate hash `3dc2a1f840b3eb725bea629dc130f070`; version `lm-ancova-v2-2026-1`; held-out not opened).
- **ANCOVA violation-detection study.** This re-aim met quotas but was **not confirmed**: pooled $\Delta\text{AUC} = \text{AUC}_{\text{score}} - \text{AUC}_{\text{p}} = 0.0053$ with scenario-cluster bootstrap 95% CI $[-0.111, 0.129]$ (the pre-specified requirement was $\Delta\text{AUC} \geq 0.10$ and CI lower bound > 0).

Registry rows for `lm_ancova` and `lm_ancova_v2` remain uncalibrated with auditable provenance; categorical labels stay suppressed. Welch 55/70 is a Welch comparator only and is never an ANCOVA fallback. The frozen `pain_ancova_trial` dataset is an illustration only and never enters training.

2.4 Statistical implementation

Implemented in the `stabilitest` R package (this repository). `robustness_analysis()` covers two-sample location tests (Welch and paired t; Wilcoxon rank-sum/signed-rank; Brunner–Munzel) and two-group binary proportion tests (`fisher`, `chisq`, `prop`) without changing the public dispatcher. Rank-based location analyses report the Hodges–Lehmann shift as the effect summary. Model engines: `robustness_lm()` (linear/ANCOVA), `robustness_glm()` (binomial logit / Poisson log), and `robustness_surv()` (Cox); each accepts a single coefficient or a multi-df factor term tested jointly (F for `lm`; LRT for `glm/surv`). Equivalence and non-inferiority use `robustness_tost()` (TOST / one-sided margin tests) for mean, risk-difference, and odds-ratio endpoints. Numeric scores and components are retained across all these engines; categorical labels are emitted only for the active `fisher_exact` calibration described in Section 2.3. Computational cost is dominated by the greedy search: $O(n \cdot k_{\max})$ test evaluations, about 3,400 tests for $n = 55$ — under a second for closed-form tests, minutes for model refits at n of a few hundred.

2.5 Application context

Primary use: two-arm comparisons of continuous endpoints, $n \approx 20$ –200 per group, Phase I–III — the setting in which sensitivity of conclusions to individual patients is a recurring concern in drug development and trial reporting (Senn, 2007; Pocock, McMurray & Collier, 2015). The row-deletion engine (Section 2.6) extends the same logic to covariate-adjusted analyses. Observations are assumed independent; clustered and longitudinal extensions are future work (Section 5).

2.6 Model-based extensions

For ANCOVA (`change ~ arm + baseline`), GLM (`y ~ arm + covariates`; binomial logit or Poisson log), and Cox regression (`Surv(time, event) ~ arm + covariates`), all components operate on *rows* (subjects) of the analysis dataset, with the conclusion defined by the p-value of a pre-specified model term — a single coefficient, or a multi-df factor term via a joint test (`drop1` F for `lm`; LRT for `glm/surv`). For Cox with a single binary arm the score test is asymptotically the log-rank test. Deleting whole subjects — with their covariates and follow-up — preserves the adjustment structure and handles censoring naturally. Model fits that fail after deletion (e.g., a factor level vanishing) are skipped and reported.

3. Historical Welch simulation evidence (descriptive)

The following simulation and case-study tables are retained to document the original broad-family experiment. They are historical descriptive evidence about Welch score and removal-index behavior, not confirmatory calibration for Welch or any other method. The current runtime registry and label policy are defined in Section 2.3 and the package calibration registry.

3.1 Design

Full factorial: effect size $d \in \{0, 0.5, 0.8\}$ (data $N(0,1)$ vs $N(d,1)$); $n \in \{25, 50\}$ per group; contamination $\in \{\text{none}, 2 \text{ outliers at } +4 \text{ SD injected into the treated group, inflating the apparent effect}\}$. 500 replications per scenario; Welch t-test at $\alpha = 0.05$; $B = 200$; $k_{\max} = 30\%$ of the pooled sample; default weights. Monte Carlo $SE \leq 0.022$ for proportions. (Runtime note: `simulation_study.R` reproduces the design; results below were generated with an algorithmically identical implementation of the same components — differences across RNGs are within Monte Carlo error.)

3.2 Rejection rates and overall score (all replications)

d	n/group	Outliers	Rejection rate	Mean score (SD)
0	25	0	0.052	67.7 (8.6)
0	25	2	0.092	65.6 (10.8)
0	50	0	0.046	64.2 (6.7)
0	50	2	0.068	63.4 (6.7)
0.5	25	0	0.424	61.5 (11.3)
0.5	25	2	0.638	61.5 (11.7)
0.5	50	0	0.728	60.9 (10.7)
0.5	50	2	0.844	63.3 (10.5)
0.8	25	0	0.774	67.8 (14.0)
0.8	25	2	0.912	71.4 (13.3)
0.8	50	0	0.976	74.9 (10.6)
0.8	50	2	0.988	77.0 (10.1)

Type I error is preserved without contamination and inflated by directional outliers (0.092 at $n = 25$), exactly the situation in which a robustness assessment is needed.

3.3 Calibration conditional on significance (Table 2)

Among significant replications:

d	n/group	Outliers	Mean score	Median k_frag (worst-case)	Median fragility %	Median k (extreme-value)	S_jack	S_boot
0	25	0	52.2	1	2.0	—	84.5	65.5
0	25	2	51.6	1	2.0	—	84.3	65.6
0	50	0	51.6	2	2.0	—	87.2	64.2
0	50	2	54.8	2	2.0	—	93.9	68.2
0.5	25	0	62.7	3	6.0	13	94.7	75.9
0.5	25	2	63.2	3	6.0	7	95.3	78.1
0.5	50	0	63.1	5	5.0	14	97.3	80.1
0.5	50	2	65.3	6	6.0	12	98.7	84.2
0.8	25	0	70.8	5	10.0	10	97.5	84.5
0.8	25	2	73.0	6	12.0	8	98.5	88.2
0.8	50	0	75.4	12	12.0	26	99.7	93.9
0.8	50	2	77.3	14	14.0	28	99.8	96.0

Three descriptive findings emerged. **First, the average score differed across the simulated truth strata:** false positives under the null averaged 52, while large-effect scenarios averaged 71–77. The distributions nevertheless overlapped too much to support categorical interpretation. In the later prospective training study, the historical 55/70 rule falsely reassured 45.8% of null results and identified only 49.9% of clear-effect results, so it failed the frozen requirements. **Second, null scenarios had a median worst-case fragility of 1–2 observations (2% of the sample), versus 10–14% for large-effect scenarios.** These are group-level empirical patterns, not diagnostic signatures for an individual trial. They mirror, at trial scale, the finding of Broderick et al. that conclusions can hinge on a small fraction of observations. **Third, symmetric outlier-trimming overstated stability:** median flipping sets under grand-mean removal were 2–5 times larger than under adversarial removal in the same data (e.g., 14 vs 5 at $d = 0.5$, $n = 50$). A result that “survives outlier removal” may still be one clever deletion away from reversal.

Jackknife stability exceeded 84% everywhere and 97% for all $n = 50$ scenarios with true effects, confirming that leave-one-out stability saturates with n and cannot serve as the primary fragility measure (Section 2.1).

4. Historical Welch case example: Phase II Analgesic Trial

This case study predates the method-specific registry. Its Welch result remains useful as a reproducible example of the numeric outputs and their clinical follow-up, but the retired 55/70 interpretation is not a current verdict.

4.1 Study and primary analysis

Randomized Phase II trial of a novel analgesic vs placebo in chronic pain; primary endpoint: change from baseline in pain score (0–100) at Week 12. The analysis dataset is shipped with the software (`pain_treatment`, $n = 28$; `pain_placebo`, $n = 27$) so that every number below is exactly reproducible (bootstrap component: $B = 2000$, seed = 14; see Appendix A).

Treatment: mean change -19.80 (SD 13.80). Placebo: -8.40 (SD 11.88). Welch t-test: difference -11.40 (95% CI -18.35 to -4.44), $t = -3.286$, $df = 52.3$, $p = 0.0018$. The treatment group includes one extreme responder (-52.0 , subject 14); the placebo group includes a small-magnitude positive change ($+3.0$, subject 22), distinct from a larger placebo worsening ($+23.0$, subject 16).

4.2 Robustness analysis

Overall numeric score: 72.5/100; no categorical Welch verdict is emitted, with weights 0.4/0.4/0.2, $B = 2000$, and seed = 14. Under `welch-2026-2`, the score remains available for transparent component reporting but has no validated Robust, Moderately Robust, or Fragile interpretation.

Jackknife: 100% conclusion stability across all 55 leave-one-out tests; p-value range 0.0006–0.0034; no observation met the influence criterion. As anticipated by the simulation, this near-perfect stability mostly reflects $n = 55$, not invulnerability.

Worst-case removal: fragility index **$k_frag = 6$** (10.9% of the sample). Greedy trajectory: $p = 0.0018 \rightarrow 0.0034 \rightarrow 0.0062 \rightarrow 0.0114 \rightarrow 0.0201 \rightarrow 0.0349 \rightarrow 0.0602$ after the sixth deletion. Six specific patients — led by the extreme responder — jointly carry the conclusion; their removal is sufficient (though not necessarily minimal) to lose significance. For descriptive context, the historical simulation produced medians of 5–6 for large-effect scenarios at a similar sample size and 1–2 for null scenarios. That comparison is not a calibrated diagnosis of the trial's truth.

Extreme-value removal (descriptive): also flips at $k = 6$ in this dataset — here the grand-mean extremes coincide with the most damaging observations, which is not guaranteed in general (Section 3.3).

Bootstrap: reproducibility probability $\approx 92\%$ ($B = 2000$, seed = 14; mean bootstrap $p = 0.019$, SD 0.065; percentile interval < 0.001 to 0.175). (Bootstrap figures vary by $\pm \sim 1$ point across RNG streams; all other numbers above are deterministic.)

4.3 Interpretation and reporting

The component profile shows strong primary evidence ($p = 0.0018$), perfect leave-one-out stability, a six-subject worst-case removal set, and a high bootstrap same-decision rate. It does not establish a validated Robust diagnosis. Two actions are warranted for the CSR: clinical review of subject 14 (protocol adherence, concomitant medication) since this patient heads the worst-case removal set; and a supplementary rank-based analysis, which is less leveraged by extreme responders.

Suggested reporting text (Results): “Robustness analysis (stabilitest v0.6.0; $B = 2000$, seed = 14) yielded a numeric score of 72.5/100; the Welch categorical label was suppressed under calibration version

welch-2026-2. All 55 leave-one-out analyses preserved statistical significance ($p \leq 0.0034$). Worst-case removal analysis identified a set of 6 patients (10.9% of the sample) whose exclusion would raise the p-value to 0.060. The bootstrap same-decision rate was 92%.”

5. Discussion

5.1 Principal findings

The framework separates three questions usually blurred together — who drives the result, how small a deletion overturns it, and would it replicate. In the historical Welch simulation, chance-significant findings scored about 52 on average and large-effect scenarios about 75, but those exploratory anchors did not satisfy the later prospective error requirements. The `welch-2026-2` training result was therefore fail-closed, while `fisher_exact` independently earned a two-band Fragile / Not fragile mapping at $L = 58$ under explicit jackknife-light weights. The three ANCOVA studies — two no-cutoff training results and one unconfirmed violation-detection re-aim — further demonstrate why mappings cannot be transferred between methods. The starkest practical lesson from the historical Welch simulation remains the gap between adversarial and symmetric removal: claims based on “we removed the outliers and the result held” can overstate stability several-fold.

5.2 Interpretation guidance

For a `welch_unpaired` result, report the numeric score and each component without assigning a categorical band. Use the worst-case removal set to guide clinical and data-quality review, show the p-value trajectory, and consider a rank-based supplementary analysis when extreme observations matter. For an applicable, significant `fisher_exact` result under explicit jackknife-light weights (`fragility = 0.5`, `bootstrap = 0.5`, `jackknife = 0`) and a canonical profile, score ≤ 58 is **Fragile** and score > 58 is **Not fragile**; there is no Robust tier for Fisher, and default 0.4/0.4/0.2 weights keep the label suppressed. Other methods and conclusions retain numeric scores and components but have suppressed categorical labels. A worst-case fragility below about 5% of the sample is a useful review prompt, not a validated diagnostic cutoff. For non-significant results, use the components descriptively (Section 2.3).

5.3 Regulatory alignment

The framework operationalizes ICH guidance on conclusion robustness. ICH E9 states that it is important to evaluate “the robustness of the results and primary conclusions of the trial,” defining robustness as “the sensitivity of the overall conclusions to various limitations of the data, assumptions, and analytic approaches to data analysis.” The E9(R1) addendum elaborates this under sensitivity analysis: inferences for an estimand should be robust to data limitations and deviations from the main estimator’s assumptions, assessed through pre-specified sensitivity analyses. Accordingly, pre-specify components and weights in the SAP (template, Appendix B), and pre-specify categorical bands only when the primary analysis matches the active Fisher $L = 58$ calibration under jackknife-light weights. For Welch and other methods, report numeric scores/components and state that the categorical label was left blank. Report component metrics in the SAR and give clinical context for the worst-case removal set in the CSR. Because all metrics are descriptive functions of one pre-specified primary analysis, no multiplicity adjustment is implied.

5.4 Limitations

- (1) The binary significance framing is inherited from the fragility concept itself; a flip from $p = 0.047$ to 0.053 is a knife-edge event, and the p-value trajectory should always accompany the index. (2) The greedy fragility index is an upper bound on the minimal overturning subset; exact minimal sets are combinato-

rially hard. (3) Adversarial removal answers “could the data support the opposite conclusion?”, which is deliberately pessimistic; it should complement, not replace, assumption-based sensitivity analyses (missing data, model form). (4) Independence of observations is assumed. (5) Active categorical calibration covers only `fisher_exact` two-band interpretation ($L = 58$ under explicit jackknife-light weights). Welch, remaining model-based, `proportion (chi_square_2x2, two_sample_prop)`, rank, Brunner–Munzel, Cox, and TOST engines keep their labels suppressed pending an independently successful calibration. (6) Weight choice remains a convention, now explicit and pre-specifiable; each method-specific calibration must be earned independently.

5.5 Future work

Independent calibration for `chi_square_2x2` (binary-proportion Phase 2) is the next positive target. The Welch and `lm_ancova` negative-result programs remain archived with suppressed labels; any further categorical attempt would require a new frozen design, not reuse of the retired Welch 55/70 anchors. Remaining priorities: GLM/Cox, rank, and TOST units; clustered and longitudinal data; exact or certified bounds on minimal overturning subsets (integer-programming formulations); Bayesian analogues (prior-sensitivity and posterior-probability fragility); CDISC ADaM integration; interactive reporting.

6. Conclusions

Statistical significance answers “is there an effect?” Robustness analysis answers “how much of the data does that answer rest on?” The revised framework quantifies both fragility (worst-case removal) and reproducibility (bootstrap), identifies the specific patients who carry the conclusion, and retains numeric scores/components across methods. Active categorical interpretation is limited to the applicable `fisher_exact` two-band configuration. The Welch and ANCOVA programs are reported as named negative results so their exploratory mappings are not reused. We recommend pre-specified application to primary endpoints of confirmatory trials, with the component metrics — not the composite alone — as the substance of reporting.

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Appendix A: Software

Complete implementation in this repository (package v0.6.0): `robustness_analysis.R` (two-sample framework — including proportion tests, Brunner–Munzel, and Hodges–Lehmann reporting — and case-study data), `robustness_models.R` (lm/ANCOVA, GLM, Cox; multi-df joint tests), `robustness_tost.R` (equivalence / non-inferiority for mean, prop, and OR endpoints), `simulation_study.R` (Section 3 design), and the `stabilitest/` R package (functions, bundled case-study data, unit tests). The Section 4 case-study numbers (overall score ≈ 72.5 ; bootstrap reproducibility $\approx 92\%$) use `n_boot = 2000` and `seed = 14` (the package default seed is 123 and yields a slightly different bootstrap component). Reproduce the analysis by:

```
library(stabilitest)
res <- robustness_analysis(pain_treatment, pain_placebo,
                          test_type = "t.test", n_boot = 2000,
                          seed = 14, interpret = TRUE)

print(res)
plot(res)
# Also: test_type = "wilcoxon" | "brunner_munzel" | "fisher" | "chisq" | "prop"
# Models: robustness_lm(), robustness_glm(), robustness_surv()
# Equivalence / NI: robustness_tost(..., endpoint = "mean" | "prop" | "or")
```

To reproduce the Section 3 simulation against the exact checkout containing the manuscript, install the development packages `pkgload` and `tidyverse`, then run the script from any working directory:

```
Rscript /path/to/stabilitest/manuscript/simulation_study.R
```

The script resolves the repository from its own location and loads that checkout with `pkgload`; it does not use a potentially stale installed copy of `stabilitest`.

Appendix B: SAP/SAR template (updated)

SAP — Sensitivity and Robustness Analyses. Robustness of the primary efficacy analysis will be assessed with the `stabilitest` framework: (1) jackknife leave-one-out analysis (influence criterion: significance flip or $|\Delta p| > 0.05$); (2) worst-case greedy removal up to 30% of the sample, yielding the removal fragility index; (3) bootstrap resampling ($B = [1000]$ iterations) estimating the reproducibility probability. The numeric composite score and component metrics will be reported for every method. A categorical band will be reported

only for an applicable Fisher exact analysis under explicit jackknife-light weights: Fragile iff score ≤ 58 ; Not fragile iff score > 58 ; there is no Robust tier. Welch and other methods without validated cutoffs will retain their numeric outputs and leave the categorical label blank. If the worst-case fragility index is below 5% of the sample, this will be treated as a review prompt: the subjects in the removal set will be reviewed for data quality and a rank-based supplementary analysis performed where appropriate.

SAR — Results skeleton. Overall numeric score [XX]/100; categorical label [band or suppressed, with calibration version and eligibility]. Jackknife: [XX]% stability; influential subjects [IDs]; leave-one-out p-range [[X], [X]]. Worst-case removal: fragility index [k] ([X]% of sample); p-value trajectory [Table/Figure]; removed subjects [IDs]. Extreme-value removal (descriptive): index [k]. Bootstrap: reproducibility [XX]% (B = [X]); bootstrap p mean [X], percentile interval [[X], [X]]. Clinical review note: [context for removal-set subjects].

End of manuscript.